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BAGONG PILIPINAS

IT 112: Web Systems and Technologies
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Final Project – Phase 1: Comprehensive Planning of Proposed Project: Combat Coders: The End of Bugs Reign

Submitted by:

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Project Proposal Title: Combat Coders: The End of Bugs Reign

Combat Coders is an interactive, game-based learning system that is designed to improve player's coding skills while also enhancing strategic and decision-making abilities. The game incorporates elements of adventure, card and path-based exploration, situational problem-solving, that creates an interactive and engaging environment where players progress by solving coding challenges. By incorporating programming and developing into a game, Combat Coders aims to make learning more enjoyable, immersive, and effective for students, beginners, and enthusiasts in coding.

System Purpose:

The primary purpose of Combat Coders is to provide a coding game environment that enhances user's programming skills and also their strategy and decision making, through an interactive and gamified experience. Instead of traditional learning methods like reading textbooks, players will engage in battles, choose and explore paths, and decide when it comes to scenarios that require them to solve coding problems correctly and manage resources wisely throughout the game.

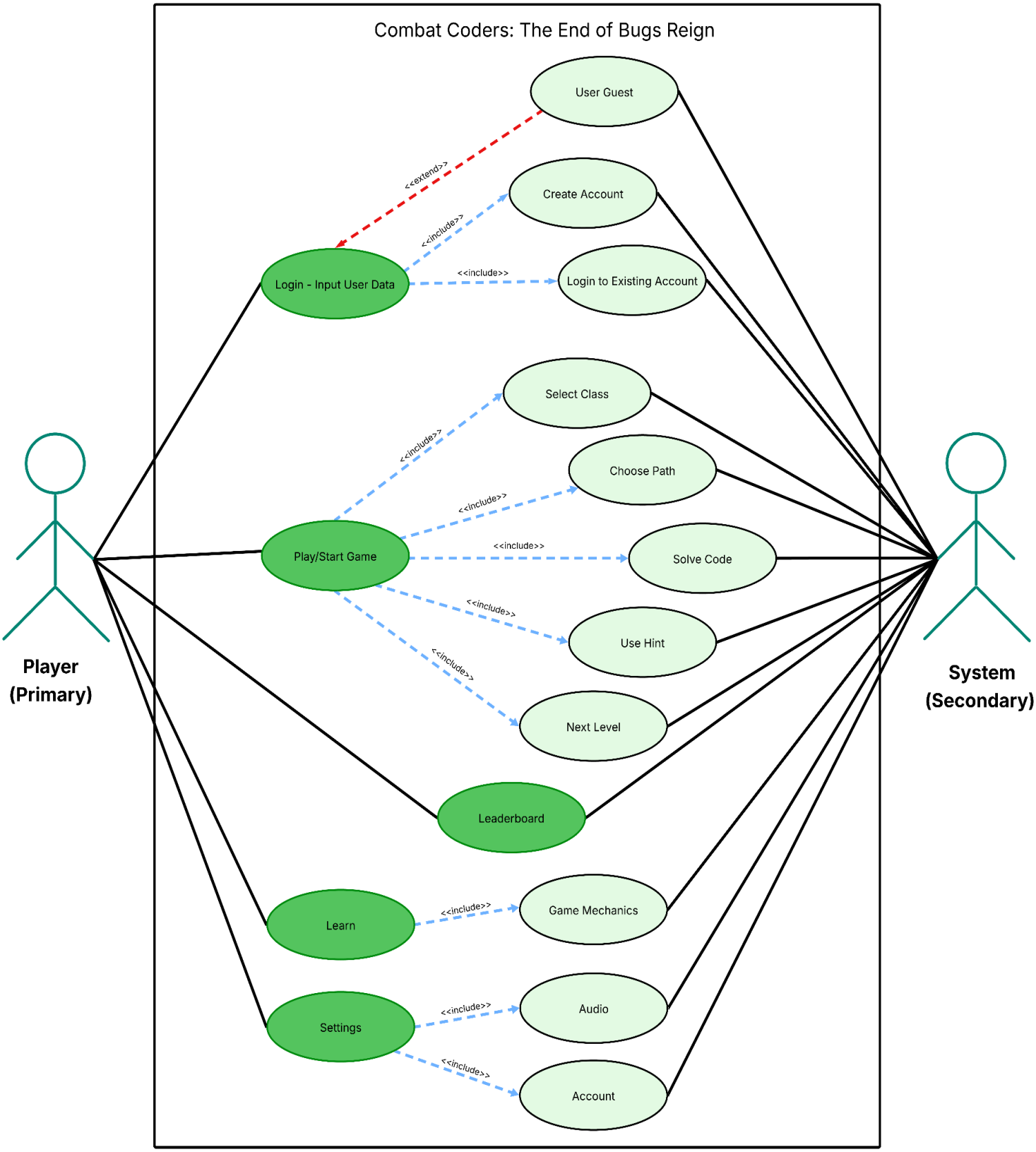
Target Users:

1. **Students:** An excellent interactive educational game activity for students in the computer studies field.
2. **Casual Gamers:** Avid players who have and do not have coding experience that are into fast-paced decision-making games.
3. **Aspiring Developers:** Game includes hands-on coding for game progression which can teach and inspire new developers.

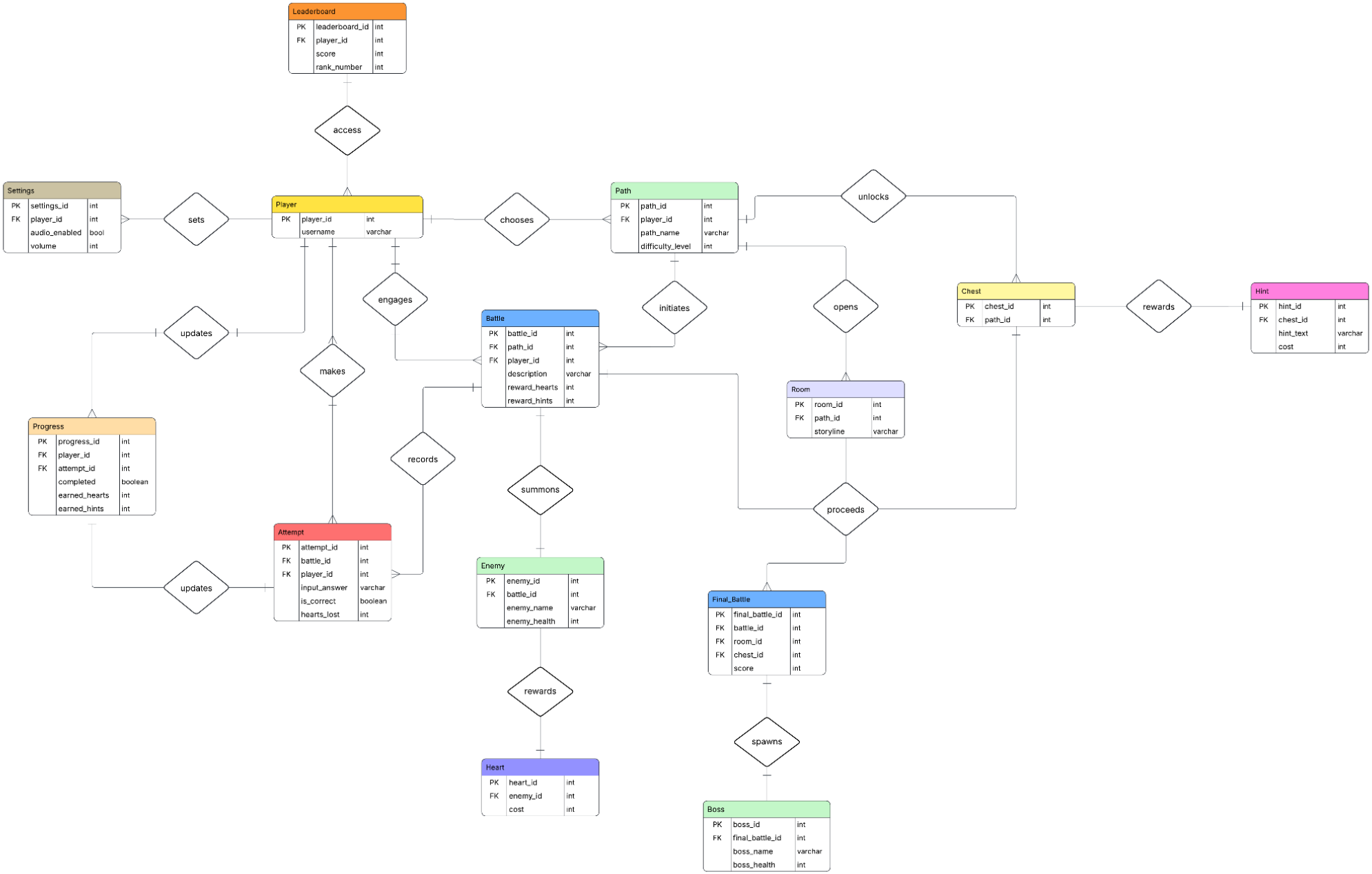
Key Features:

- **Path-Based Gameplay System:** Players navigate using a card-driven system (similar to tarot cards), where each card leads to different paths and scenarios like treasure rooms, battles, or special events, encouraging strategic decision-making.
- **Coding Battle Mechanic:** This system utilizes coding tasks as a primary combat interface. Right solutions deal damage against opponents, while wrong ones reduce the player's health (hearts).
- **Health and Reward System:** The player gains or loses health (hearts) based on performance. Winning a match grants one heart (max of 3, does not stack) and rewards called "hints," which can be used on difficult levels.
- **Progression and Level System:** Completing a level advances them further into the game. For each level completed the difficulty increases, where it introduces much more complex coding problems and strategic choices.
- **Decision-Making and Strategy Elements:** The player must choose between a chest, a room, or engage into battle. For each choice it affects the players progress, resources, and survival, enhancing their critical thinking skills.

Use Case Diagram

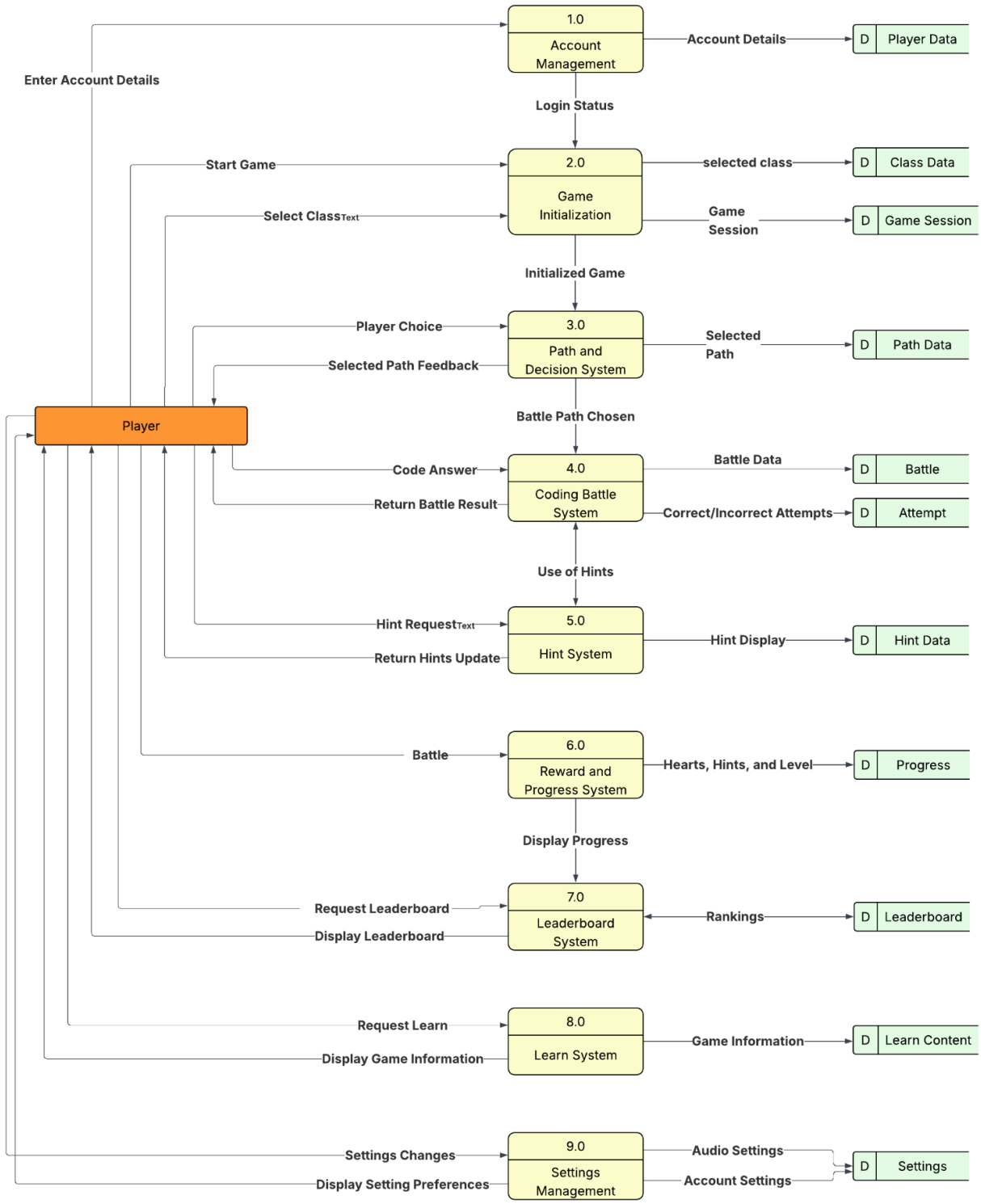


Entity-Relationship Diagram

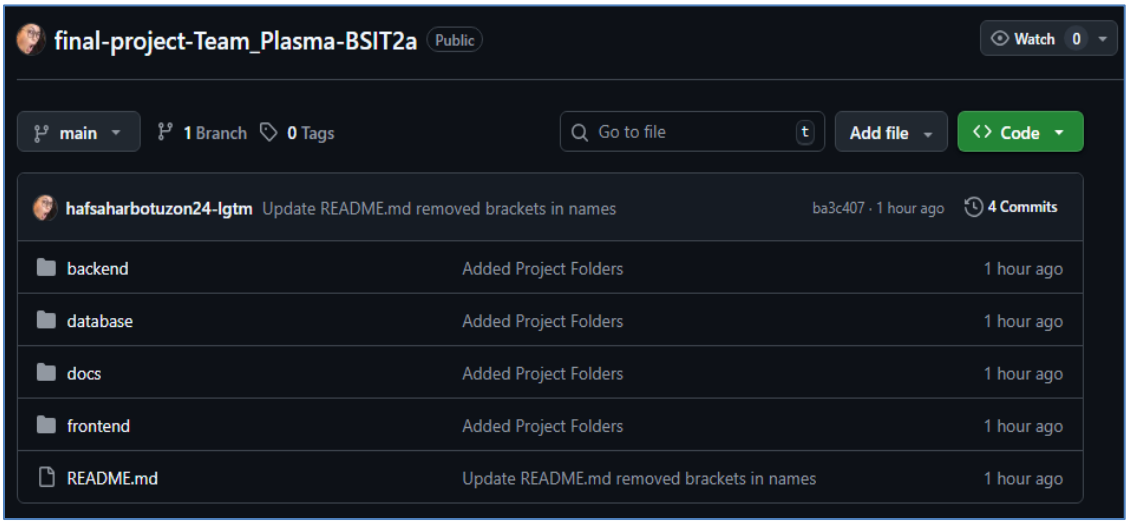
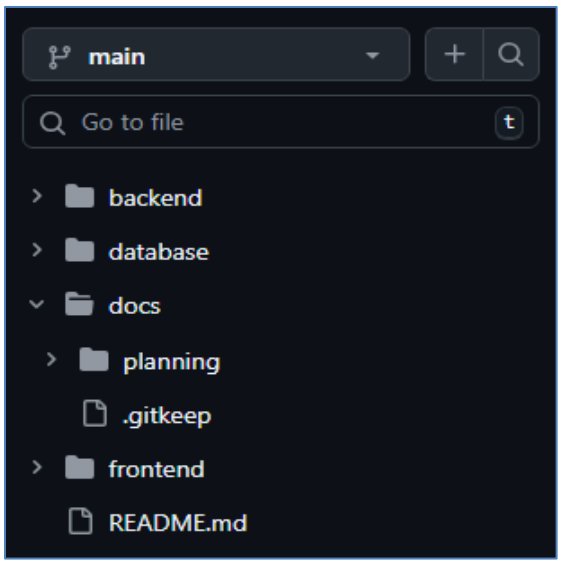


Data Flow Diagram

LVL 1 DFD



GitHub Repository URL

https://github.com/hafsaharbotuzon24-lgtm/final-project-Team_Plasma-BSIT2a

Screenshot of Repo Structure